

Master Dissertation

Stock Index Forecasting Using Tensor-Values Predictors

– Empirical Insights from the STOXX 50 Index

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Abstract

In financial markets, factor models have become classical tools for predicting asset returns. While traditional models typically focus on scenarios where the common factor is known, this dissertation delves into models where the factor remains latent, often employed for dimensionality reduction. Traditional approaches generally consider univariate time series across multiple entities (i.e., two-dimensional tensor structures). In contrast, this study explores multivariate time series across multiple entities (i.e., three-dimensional tensor structures), where data at each time point is matrix-valued.

Building upon traditional three-dimensional tensor factor models and Tucker decomposition, this dissertation introduces a novel method termed "Main Effect Decomposition" and integrates it with conventional techniques. Specifically, the method is applied to each matrix-valued slice of a three-dimensional tensor dataset to extract time-varying grand means, row, and column main effects, thereby reducing data noise. Subsequently, high-dimensional Tucker decomposition is employed to obtain more comprehensive factors, facilitating data processing and dimensionality reduction.

The effectiveness of the proposed model is validated through both simulation and empirical analysis. Simulation results demonstrate that the combination of Main Effect Decomposition with Tucker decomposition yields superior estimations, regardless of the prominence of the main effects in the data. Empirically, the innovative methodology is applied to stock index forecasting within the Lasso regression framework, incorporating tensor-valued predictors. The results indicate that this novel approach significantly enhances the accuracy of model predictions. The dissertation concludes with suggestions for future research directions and potential improvements.

Keywords: Latent factor model, main effect decomposition, Tucker decomposition, High-dimensional tensor data, index forecasting

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1 Background and Introduction

With the advent of the information age, data has become an essential component of our daily lives, serving as a key carrier of information. Data comes in various forms, encompassing different dimensions of time, space, fields, and levels, each contributing to diverse structures. One such structure, capable of representing complex information and intricate relationships, is known as a tensor. A tensor is a mathematical concept that refers to a multidimensional array. Figure 1 provides a basic definition of tensors. A zero-order tensor is a scalar, a

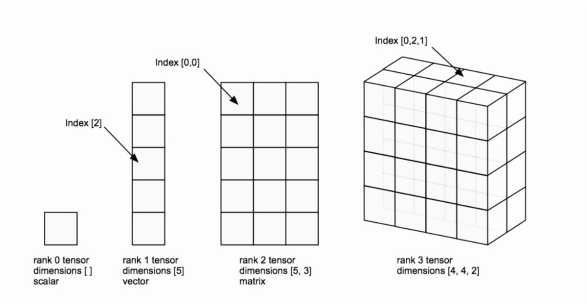


Figure 1: Tensor diagram

first-order tensor is a vector, a second-order tensor is a matrix, and a third-order tensor takes a cubic form. Tensors of order three or higher are referred to as higher-order tensors. Formally, an N-th-order tensor is an element of the tensor product of N vector spaces. Tensors effectively preserve the spatial structure and multidimensional information of the original data, making them a prominent focus in contemporary statistical research and analysis. This

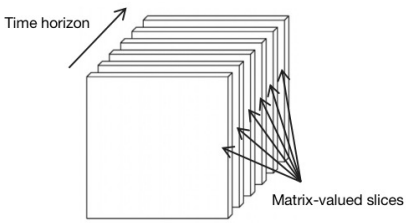


Figure 2: Data structure

study primarily examines second-order tensors (matrix-valued data) and third-order tensors. Specifically, at each time point, we have matrix-valued data, which is aggregated over time to create a cubic data structure, as illustrated in Figure 2.

Compared to low-dimensional tensors, high-dimensional tensors typically carry more information. However, the increase in dimensions introduces challenges such as correlation and redundancy, known as the "Curse of Dimensionality," first proposed by Bellman (1961).

For instance, according to Lam et al. (2011), the standard multiple time series models such as vector autoregressive or vector autoregressive-moving-average models are rarely used in practice without dimension reduction or regularization, since when the number of time series N is high, the number of parameters involved is of the order of N^2.

To address these challenges and achieve dimension reduction, tensor decomposition methods have been developed. Decomposition techniques are used to extract and interpret the properties of tensors and have been a significant area of research over the past decades. The concept of tensor decomposition was first introduced by Hitchcock (1927), and the idea of a multiway model was advanced by Cattell (1944). In Section 2 Literature Review, we will begin by reviewing the traditional factor models for dimension reduction and their extension to third-order tensors. This section will also cover recent advancements in the application of tensor data and factor models. Additionally, we will introduce the fundamental principles of constrained regression for forecasting and validation. In Section 3 Methodology, we present a novel decomposition method called main effect decomposition for two-dimensional data. This method will be used to extract time-varying grand means, row effects, and column effects of matrix-valued data at each time point, before applying further classic decomposition methods to the entire three-dimensional tensor. In Section 4 Simulation Analysis, we will test our hypothesis that the main effect decomposition enhances the performance of traditional methods and provides better estimates of data.

While establishing a solid theoretical foundation is crucial, it is equally important to apply these concepts in practical scenarios. Previous research has demonstrated various applications of tensor decomposition across different fields. For instance, works by Vasilescu and Terzopoulos (2002), Shashua and Hazan (2005), and Hazan et al. (2005) have explored tensor decomposition in computer vision, while Acar et al. (2007), Beckmann and Smith (2005), and De Vos et al. (2007) have applied these methods in neuroscience. Additionally, Chen et al. (2002), Comon (2002), and De Lathauwer and Vandewalle (2004) have investigated tensor decomposition in signal processing. In this research, as detailed in Section 5 Empirical Study, we apply both traditional tensor decomposition methods and our newly proposed main effect decomposition within the context of financial markets. We will evaluate the effectiveness and accuracy of the new method through empirical analysis. Specifically, we will use tensor-valued time series data, which includes stock characteristics from a portfolio at various time points, to predict market index movements. The key contribution of this paper is the development of tensor-valued predictors based on both main effect decomposition and traditional methods. Ultimately, we will compare the predictive accuracy of the new and

traditional models to assess their performance.

In summary, this paper focuses on tensor data, utilizing both simulated data and real financial market data to evaluate the effectiveness of the proposed new model and method. This approach aims to address and fill the existing research gaps in the current literature.

2 Literature Review

High-dimensional time series analysis is widely used in finance and economics. For example, analyzing the dynamics of returns across numerous assets is essential for asset pricing, portfolio allocation, and risk management. To address the complexities of high-dimensional data, factor modeling is a commonly employed method for dimension reduction when analyzing multiple time series (Lam et al., 2011).

The basic factor model is expressed as follows:

$$r_{it} = \alpha_i + \boldsymbol{\beta}_i^T \boldsymbol{f}_t + \epsilon_{it} \tag{2.1}$$

where $\boldsymbol{f}_t = (f_{t1}, ..., f_{tK})^T$ is a vector of K factors and $\boldsymbol{\beta}_i = (\beta_1, ..., \beta_K)^T$ is the corresponding coefficient vector for identity i at time t; r_{it} denotes the observed value for identity i at time t; and ϵ_{it} represents the noise term. In matrix notation, this model can be written as:

$$\boldsymbol{r}_t = \boldsymbol{\alpha} + \boldsymbol{B}\boldsymbol{f}_t + \boldsymbol{\epsilon}_t \tag{2.2}$$

where $\boldsymbol{B} = (\boldsymbol{\beta}_1, ..., \boldsymbol{\beta}_p)^T$ represents the factor loading matrix, where each $\boldsymbol{\beta}_i \in \mathbb{R}^{K\times 1}$ with $i \in \{1, 2, ..., K\}$. The $\boldsymbol{\beta}_i$ serves as regression parameter for the i-th identity when the factors $\boldsymbol{f}_t$ is known. In this context, the Equation 2.2 is referred to as a "known factor model." Notable examples include the Capital Asset Pricing Model (CAPM), developed independently by Sharpe (1964), Lintner (1965), and Mossin (1966) with the number of factors $K = 1$ and the Fama-French three-factor model proposed by Fama and French (1993) with with the number of factors $K = 3$, representing three common risk factors. When the factors $\boldsymbol{f}_t$ are unknown, Equation 2.2 is considered a latent factor model, which aims to reduce dimensionality by extracting common factors from multiple interrelated variables. This paper focuses on latent factor models. Early theoretical work in this area includes studies by Priestley et al. (1974) and Brillinger (2001). Empirically, researchers have explored methods such as obtaining indirect measurements of latent variables in psychological statistics Anderson (1963) and identifying hidden factors in multivariate time series processes. For instance, Pena and Box (1987) examined wheat prices in five provinces of Spain during the 19th century as an illustrative example. The latent factor model also has significant implications in finance and economics fields. Chamberlain (1983) introduced a less restrictive definition of factor structure compared to traditional arbitrage pricing models, while Stock and Watson (2002) utilized principal component analysis (PCA) to extract common factors, specifically diffusion

indices, from high-dimensional macroeconomic time series data to forecast inflation. Overall, factor analysis for univariate time series, where a single variable is observed over time, is well-established in the previous studies (Wang et al., 2019).

Next, we extend the model to higher dimensions. The previously discussed factor model mainly deals with matrix data, or two-dimensional tensors, which represent data for the same variable across different individuals and time points. We now consider cubic data, or three-dimensional tensors, which encompass data for multiple variables across different individuals at various time points, where each time point is represented by a matrix. While high-dimensional data generally provides more information than low-dimensional data, it also presents significant challenges due to correlations and redundancies. As a result, matrix-valued time series have received considerable attention in recent years for dimension reduction and predictive modeling across various fields, including finance, economics, medical science, and meteorology (Lam and Cen, 2024). Panel time series—where multiple variables are observed over time—has been a well-established and actively researched area, as noted by Stock and Watson (2005), Lütkepohl (2005), and Tsay (2013).

Before we delve deep into tensor decomposition for dimension reduction, let's review the foundation and basic calculations for tensors. The Hadamard product of two tensors $\mathcal{A}$ and $\mathcal{B}$ is a new tensor obtained by element-wise multiplication of $\mathcal{A}$ and $\mathcal{B}$. For example, if tensor $\mathcal{A} \in \mathbb{C}^{I_1 \times I_2 \times I_1}$ and tensor $\mathcal{B} \in \mathbb{C}^{I_1 \times I_2 \times I_2}$, their Hadamard product $\mathcal{C} \in \mathbb{C}^{I_1 \times I_2 \times I_1 \times I_2}$ is represented as $\mathcal{A} * \mathcal{B} = \mathcal{C}$. Additionally, the Kronecker product for matrices $A \in \mathbb{C}^{I \times J}$ and $B \in \mathbb{C}^{M \times N}$ is defined as a large matrix of size $IM \times JN$, expressed as $A \otimes B$. The operation is detailed by the equation $A \otimes B = [a_1 \otimes b_1, a_1 \otimes b_2, \ldots, a_J \otimes b_N]$, where a_j with $1 \leq j \leq J$ and b_n with $1 \leq n \leq N$ are column vectors of matrices A and B respectively. Moreover, the Khatri-Rao product, introduced by Khatri and Rao (1968), also known as the column-wise Kronecker product, is specific to the multiplication of corresponding columns of two matrices. For given matrices $A \in \mathbb{C}^{I \times K}$ and $B \in \mathbb{C}^{J \times K}$, the result is a matrix of size $IJ \times K$, denoted as $A \odot B$ and represented by $A \odot B = [a_1 \otimes b_1, a_2 \otimes b_2, \ldots, a_K \otimes b_K]$. These fundamental tensor operations are crucial in understanding more complex tensor manipulations and decompositions.

As a prominent area in statistical modeling and parameter estimation, tensor data have well-established theories and algorithms for decomposition, such as CP (CANDECOMP / PARAFAC) decomposition and Tucker decomposition. Initially, Hitchcock (1927) introduced the concept of the polyadic form of a tensor. Subsequently, Cattell (1944) proposed a multi-dimensional tensor decomposition model, elaborating on parallel proportional analysis. This method gained prominence and recognition in the field of psychological testing. Carroll and

Chang (1970) introduced the Canonical Decomposition (CANDECOMP) method, which, along with the Parallel Factors (PARAFAC) algorithm (Harshman, 1970), is collectively known as CP decomposition. As illustrated in Figure 1, the CP decomposition factorizes

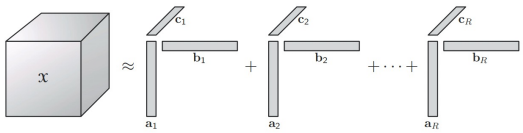


Figure 3: CP decomposition illustration

a tensor into a sum of component rank-one tensors. For example, for a third-order tensor $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{I \times J \times K}$, it can be expressed as:

$$\boldsymbol{\mathcal{X}} \approx \sum_{r=1}^{R} \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r \tag{2.3}$$

where R is a positive integer and $\mathbf{a}_r \in \mathbb{R}^I$, $\mathbf{b}_r \in \mathbb{R}^J$, and $\mathbf{c}_r \in \mathbb{R}^K$ for $r = 1, \ldots, R$. Tucker decomposition, first introduced by Tucker (1963) and refined in subsequent studies such as Levin (1965), is further elaborated in the comprehensive work by Tucker (1966), who considered the application of tensors in factor analysis. As shown in Figure 2, Tucker decomposition

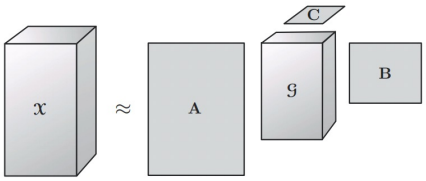


Figure 4: Tucker decomposition illustration

can be viewed as a higher-order extension of PCA (Kolda and Bader, 2009). It decomposes a tensor into a core tensor multiplied by matrices along each mode. For a three-way tensor $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{I \times J \times K}$, this is expressed as:

$$\boldsymbol{\mathcal{X}} \approx \boldsymbol{\mathcal{G}} \times_1 \mathbf{A} \times_2 \mathbf{B} \times_3 \mathbf{C} = \sum_{p=1}^{P} \sum_{q=1}^{Q} \sum_{r=1}^{R} g_{pqr} : \mathbf{a}_p \circ \mathbf{b}_q \circ \mathbf{c}_r \tag{2.4}$$

where $\mathbf{A} \in \mathbb{R}^{I \times P}$, $\mathbf{B} \in \mathbb{R}^{J \times Q}$, and $\mathbf{C} \in \mathbb{R}^{K \times R}$ are factor matrices (typically orthogonal) that can be seen as the principal components in each mode. The tensor $\boldsymbol{\mathcal{G}} \in \mathbb{R}^{P \times Q \times R}$, known as the core tensor, captures the interaction between the different components. Compared to CP decomposition, Tucker decomposition includes a core tensor that retains more information from the original tensor data, allowing for better reconstruction and making it more broadly

applicable in practice.

In this study, we focus on three-dimensional tensor data by preserving the time dimension and applying dimensionality reduction to the remaining two dimensions. This approach transforms our analysis from a global cubic tensor into a time series comprising two-dimensional matrices at each time point. For matrix-valued time series, Tucker decomposition can be expressed as:

$$\boldsymbol{X}_t = \boldsymbol{\alpha}_t + \boldsymbol{R}\boldsymbol{f}_t\boldsymbol{C}^T + \boldsymbol{\epsilon}_t \tag{2.5}$$

where $\boldsymbol{X}_t \in \mathbb{R}^{M\times N}$ represents the observed matrix at time t, with M rows and N columns; $\boldsymbol{R} \in \mathbb{R}^{M\times D_r}$ and $\boldsymbol{C} \in \mathbb{R}^{N\times D_c}$ are the row and column factor loading matrices, respectively; $\boldsymbol{f}_t \in \mathbb{R}^{D_r\times D_c}$ is the core factor term with D_r rows and D_c columns; and $\boldsymbol{\epsilon}_t \in \mathbb{R}^{M\times N}$ is the noise term at time t. Significant theoretical and methodological advancements have been made in the field of matrix-valued time series. For instance, Wang et al. (2019) developed a novel factor model that leverages the matrix structure to achieve enhanced dimensionality reduction and clearer, more interpretable factor structures. Chen et al. (2020) introduced a general framework that integrates domain and prior knowledge into matrix factor models through linear constraints. He et al. (2024) proposed a large-dimensional matrix factor model, estimating the factor loading matrices and factor score matrix by minimizing a square loss function. These studies focus on matrix-valued factor models using Tucker decomposition for common components. In contrast, Chang et al. (2023) proposed a new, one-pass estimation procedure based on generalized eigenanalysis, derived from the serial dependence structure of the underlying process, for CP decomposition of common components.

Based on extensive studies in this field, this research proposes a new method called main effect decomposition, presented in Equation 3.2, with specific constraints detailed in Equation 3.3, which will be further explained in Section 3 Methodology. This method applies to matrix-valued data to derive the time-varying grand mean, row and column main effects, and the interaction term. Subsequently, the interaction term can be decomposed as shown in Equation 3.11, which is the same method with Equation 2.5. The main effect decomposition is a novel model designed to extract the main effects based on the row and column features of matrices. It addresses a research gap by reducing noise in the original data, thus enhancing prediction accuracy and model interpretability.

After clarifying the concepts of main effect decomposition and the latent factor model, the next crucial step is to determine the number of factors, a prominent challenge in the high-dimensional field. Numerous classical studies have addressed this issue. Bartlett (1937) initially proposed a method using likelihood ratio tests to determine the number of factors.

Later, Cattell (1966) introduced the scree plot method, which involves analyzing the trend of changes in matrix eigenvalues to identify the number of factors. However, interpreting the scree plot can be subjective, especially when the plot lacks a clear elbow or shows multiple elbows. Gorsuch (1973) expanded on Bartlett's approach with an extended application of the likelihood ratio test for factor determination. Humphreys and Montanelli (1975) introduced Parallel Analysis, a widely used method in exploratory factor analysis for deciding the number of factors. Bai and Ng (2002) proposed using information criteria functions (AIC and BIC) to determine the number of factors. As factor model theory has advanced, new methods such as ratio-based estimators have gained prominence. Lam and Yao (2012) and Ahn and Horenstein (2013) developed estimators based on eigenvalue ratios, while Zhang et al. (2024) proposed an estimator based on cumulative eigenvalue ratios. In this research, we utilize eigenvalue-ratio estimators to determine the dimension of the core factor $\boldsymbol{f}_t$ in the Tucker decomposition. These estimators are based on the row and column sample covariance matrices, which are estimated from the interaction terms obtained during the main effect decomposition process for each time period. Further details on this approach will be provided in Section 3 Methodology.

After determining the number of factors using the main effect decomposition proposed in this research and the traditional Tucker decomposition, we obtain the time-varying row and column effects, as well as the core factor with a specified dimension. Importantly, we maintain the original time dimension throughout this process. Next, we vectorize the row and column effects and the core factor for each time point, combining them as independent variables for subsequent regression and prediction analyses. This vectorization, denoted as $\text{vec}(\mathcal{X})$ for a tensor $\mathcal{X}$, facilitates tensor computations using vectorized operations, enables the application of tensor data in algorithms designed for vectors, and allows efficient storage and retrieval of tensor data in linear memory (Ji et al., 2019).

With the independent variables prepared, the next step involves feature selection and prediction. Traditional methods such as stepwise regression and best subset selection are commonly used for variable selection but have notable drawbacks, including high computational costs and instability, particularly when dealing with a large number of covariates. To address these issues, statisticians have developed regularization methods that optimize an objective function combining a "loss" component with a "penalty" component. These methods apply a larger penalty to smaller coefficients, effectively shrinking less important parameters to zero and thereby selecting significant variables. In the context of high-dimensional sparse variable selection, Tibshirani (1996) proposed the Lasso model, which addresses multicollinearity by

enforcing a penalty on the sum of the absolute values of the coefficients, constraining it to a fixed value. Given its ability to handle high-dimensional data and select important variables efficiently, we adopt Lasso regression as our predictive model in this research.

3 Methodology

In this section, we will present and derive the main models and methods used in this study. To begin, we will define the notation employed throughout the paper. The primary objective of this research is to use the independent variable data from time 1 to T to predict the dependent variable data from time 2 to $T+1$. At each time $t \in \{1, \ldots, T\}$, we collect N feature values for M individuals. For the n-th feature with $n \in \{1, \ldots, N\}$ of the m-th individual with $m \in \{1, \ldots, M\}$ at time t, we denote it as $x_{t,mn}$. By aggregating the independent variable data at time t, we obtain a matrix $\mathbf{X}_t$ with dimensions $M \times N$. Correspondingly, for the dependent variable at each time point t, we denote it as y_t. When combined, these form a vector $\mathbf{y}$ of length T.

Next, in Section 3.1 and Section 3.2, we will outline the methods for processing the matrix-valued independent variables $\mathbf{X}_t$. Subsequently, in Section 3.3, we will present the forecasting model used in this study.

3.1 Main effect decomposition

To begin with, we aim to decompose the data into a time-varying grand mean, row effects, column effects, and the interaction term for each time point. The basic form can be expressed as:

$$x_{t,mn} = \mu_t + \alpha_{t,m} + \beta_{t,n} + \gamma_{t,mn} + \epsilon_{t,mn} \tag{3.1}$$

where μ_t is the grand mean, $\alpha_{t,m}$ is the row effect for the m-th row, and $\beta_{t,n}$ is the column effect for the n-th column at time point t. Aggregating all elements, the formula in matrix form can be written as:

$$\mathbf{X}_t = \mu_t \mathbf{1}_M \mathbf{1}_N^T + \boldsymbol{\alpha}_t \mathbf{1}_N^T + \mathbf{1}_M \boldsymbol{\beta}_t^T + \boldsymbol{\gamma}_t + \boldsymbol{\epsilon}_t \tag{3.2}$$

where $\boldsymbol{\gamma}_t$ and $\boldsymbol{\epsilon}_t$ are $M \times N$ matrices with elements $\gamma_{t,mn}$ and $\epsilon_{t,mn}$ in the m-th row and n-th column, respectively; $\boldsymbol{\alpha}_t$ is a vector consisting of $(\alpha_{t,1}, \alpha_{t,2}, \ldots, \alpha_{t,M})^T$, and $\boldsymbol{\beta}_t$ is a vector consisting of $(\beta_{t,1}, \beta_{t,2}, \ldots, \beta_{t,N})^T$; $\mathbf{1}_M$ and $\mathbf{1}_N$ are vectors with dimensions $M \times 1$ and $N \times 1$, respectively. Additionally, for Equation 3.2, we set the following constraints to solve for the estimators:

$$\sum_{m \in M} \alpha_{t,m} = \sum_{n \in N} \beta_{t,n} = \sum_{m \in M} \gamma_{t,mn} = \sum_{n \in N} \gamma_{t,mn} = 0 \tag{3.3}$$

Here, the sums of the elements in the row effect vector $\boldsymbol{\alpha}_t$ and column effect vector $\boldsymbol{\beta}_t$ are set to zero, and similarly, the sums of each row and column in the matrix-valued interaction

term $\boldsymbol{\gamma}_t$ are also constrained to zero.

Next, combining the target function in Equation 3.2 with the constraint in Equation 3.3, we can estimate the parameters μ_t, $\boldsymbol{\alpha}_t$, $\boldsymbol{\beta}_t$, and $\boldsymbol{\gamma}_t$ using the sample data. First, for the grand mean estimator $\hat{\mu}_t$, left-multiplying Equation 3.2 by $\mathbf{1}_M^T$ and right-multiplying by $\mathbf{1}_N$ yields:

$$\mathbf{1}_M^T \mathbf{X}_t \mathbf{1}_N = \mu_t \mathbf{1}_M^T \mathbf{1}_M \mathbf{1}_N^T \mathbf{1}_N + \mathbf{1}_M^T \boldsymbol{\alpha}_t \mathbf{1}_N^T \mathbf{1}_N + \mathbf{1}_M^T \mathbf{1}_M \boldsymbol{\beta}_t^T \mathbf{1}_N + \mathbf{1}_M^T \boldsymbol{\gamma}_t \mathbf{1}_N + \mathbf{1}_M^T \boldsymbol{\epsilon}_t \mathbf{1}_N \tag{3.4}$$

where $\mathbf{1}_M^T \boldsymbol{\alpha}_t \mathbf{1}_N^T \mathbf{1}_N = \mathbf{1}_M^T \mathbf{1}_M \boldsymbol{\beta}_t^T \mathbf{1}_N = \mathbf{1}_M^T \boldsymbol{\gamma}_t \mathbf{1}_N = 0$, as stated in Equation 3.3. Therefore, the estimated grand mean is:

$$\hat{\mu}_t := \frac{\mathbf{1}_M^T \mathbf{X}_t \mathbf{1}_N}{MN} \tag{3.5}$$

For the row effect, right-multiplying Equation 3.2 by $\mathbf{1}_N$ gives:

$$\mathbf{X}_t \mathbf{1}_N = \mu_t \mathbf{1}_M \mathbf{1}_N^T \mathbf{1}_N + \boldsymbol{\alpha}_t \mathbf{1}_N^T \mathbf{1}_N + \mathbf{1}_M \boldsymbol{\beta}_t^T \mathbf{1}_N + \boldsymbol{\gamma}_t \mathbf{1}_N + \boldsymbol{\epsilon}_t \mathbf{1}_N \tag{3.6}$$

where $\mathbf{1}_M \boldsymbol{\beta}_t^T \mathbf{1}_N = \boldsymbol{\gamma}_t \mathbf{1}_N = \mathbf{0}$. Thus, the estimated row effect is:

$$\hat{\boldsymbol{\alpha}}_t := \frac{\mathbf{X}_t \mathbf{1}_N - N\hat{\mu}_t \mathbf{1}_M}{N} \tag{3.7}$$

Similarly, left-multiplying Equation 3.2 by $\mathbf{1}_M^T$ results in:

$$\mathbf{1}_M^T \mathbf{X}_t = \mu_t \mathbf{1}_M^T \mathbf{1}_M \mathbf{1}_N^T + \mathbf{1}_M^T \boldsymbol{\alpha}_t \mathbf{1}_N^T + \mathbf{1}_M^T \mathbf{1}_M \boldsymbol{\beta}_t^T + \mathbf{1}_M^T \boldsymbol{\gamma}_t + \mathbf{1}_M^T \boldsymbol{\epsilon}_t \tag{3.8}$$

where $\mathbf{1}_M^T \boldsymbol{\alpha}_t \mathbf{1}_N^T = \mathbf{1}_M^T \boldsymbol{\gamma}_t = \mathbf{0}^T$. Therefore, the estimated column effect is:

$$\hat{\boldsymbol{\beta}}_t^T := \frac{\mathbf{1}_M^T \mathbf{X}_t - M\hat{\mu}_t \mathbf{1}_N^T}{M} \tag{3.9}$$

Finally, the estimated interaction term $\hat{\boldsymbol{\gamma}}_t$ is:

$$\hat{\boldsymbol{\gamma}}_t := \mathbf{X}_t - \hat{\mu}_t \mathbf{1}_M \mathbf{1}_N^T - \hat{\boldsymbol{\alpha}}_t \mathbf{1}_N^T - \mathbf{1}_M \hat{\boldsymbol{\beta}}_t^T \tag{3.10}$$

3.2 Latent factor model

Furthermore, by obtaining the interaction term $\boldsymbol{\gamma}_t$ at each time point t through the main effect decomposition, we can then apply Tucker decomposition to it while keeping the time dimension unchanged. The matrix-valued latent factor model can be expressed as:

$$\boldsymbol{\gamma}_t = \boldsymbol{\mu} + \mathbf{R}\mathbf{F}_t\mathbf{C}^T + \mathbf{e}_t \tag{3.11}$$

where $\mathbf{R}$ is the row loading matrix with dimensions $M \times D_r$, $\mathbf{C}$ is the column loading matrix with dimensions $N \times D_c$; $\mathbf{F}_t$ is the latent factor term with dimensions $D_r \times D_c$; and $\boldsymbol{\mu}$ and $\mathbf{e}_t$ are the constant and error terms, respectively, both with dimensions $M \times N$.

Next, we will determine the row and column dimensions, D_r and D_c, for the latent factor term $\mathbf{F}_t$ as shown in Equation 3.11. According to Lam and Cen (2024), $\hat{\boldsymbol{\gamma}}_t \hat{\boldsymbol{\gamma}}_t^T$ has $\mathbf{1}_M$ in its null space, while $\hat{\boldsymbol{\gamma}}_t^T \hat{\boldsymbol{\gamma}}_t$ has $\mathbf{1}_N$ in its null space, where $\hat{\boldsymbol{\gamma}}_t$ is the estimated interaction term from Equation 3.10. Consequently, $\frac{\sum_{t=1}^{T} \hat{\boldsymbol{\gamma}}_t \hat{\boldsymbol{\gamma}}_t^T}{T}$ represents the row sample covariance matrix, and $\frac{\sum_{t=1}^{T} \hat{\boldsymbol{\gamma}}_t^T \hat{\boldsymbol{\gamma}}_t}{T}$ represents the column sample covariance matrix over the time period T. For further analysis, we compute the eigenvalues of these two sample covariance matrices separately and arrange them in descending order. This yields two sequences of eigenvalues: $\lambda_{r,1} \geq \lambda_{r,2} \geq \lambda_{r,3} \geq \cdots \geq \lambda_{r,M}$ and $\lambda_{c,1} \geq \lambda_{c,2} \geq \lambda_{c,3} \geq \cdots \geq \lambda_{c,N}$. Following Lam and Yao (2012), we then use eigenvalue-ratio estimators to determine the number of factors as follows:

$$\hat{D}_r := \arg \min_{m \in [1,M]} \frac{\lambda_{r,m+1}}{\lambda_{r,m}} \tag{3.12}$$

$$\hat{D}_c := \arg \min_{n \in [1,N]} \frac{\lambda_{c,n+1}}{\lambda_{c,n}} \tag{3.13}$$

where estimators in Equation 3.12 and Equation 3.13 are consistent and unaffected by adjustments to other complex parameters.

Overall, for a given period from 1 to T, we can estimate the dimension of the latent factor term. The next step is to apply Tucker decomposition to the tensor data. Tucker (1966) introduced three methods for computing Tucker decomposition. The first method, which is also the default method in the Python *tensorly* package, is outlined in Table 1. The core idea is to identify components that best capture the variation in the data. For a given tensor $\boldsymbol{\mathcal{X}}$, we can generally obtain an exact Tucker decomposition of rank $(R_1, R_2, \ldots, R_N)$, where $R_n = \text{rank}_n(\boldsymbol{\mathcal{X}})$. This method is now widely known as the higher-order SVD (HOSVD) (De Lathauwer et al., 2000), which is recognized as a natural generalization of the matrix SVD.

Table 1: Tucker decomposition algorithm

High-Order SVD (HOSVD) Algorithm
Procedure HOSVD($\boldsymbol{\mathcal{X}}, R_1, R_2, \ldots, R_N$)
for $n = 1, \ldots, N$ **do**
$\mathbf{A}^{(n)} \leftarrow R_n$ *(leading left singular vectors of* $\boldsymbol{X}_{(n)}$*)*
end for
$\boldsymbol{\mathcal{G}} \leftarrow \boldsymbol{\mathcal{X}} \times_1 \mathbf{A}^{(1)T} \times_2 \mathbf{A}^{(2)T} \ldots \times_N \mathbf{A}^{(N)T}$
return $\boldsymbol{\mathcal{G}}, \mathbf{A}^{(1)}, \mathbf{A}^{(2)}, \ldots, \mathbf{A}^{(N)}$
end procedure

In this research, using the Tucker decomposition algorithm with the dimension parameter $\hat{D}_r \times \hat{D}_c \times T$, we can reduce the core factor term from its original dimension of $M \times N \times T$.

3.3 Constrained regression

In this section, we will model the existing data and make predictions using three different data processing methods: raw data, Tucker decomposition (as shown in Equation 2.2), and a combined approach involving main effect decomposition and Tucker decomposition (as presented in Equation 3.2 and Equation 3.11). We will evaluate the performance of these methods by comparing their training and test errors. Regardless of the processing method, the first step involves vectorizing the data at each time point t.

Generally, following the formal definition provided by Ji et al. (2019), for a tensor $\mathcal{T}$ of order N with dimensions $d_1 \times d_2 \times \cdots \times d_N$, tensor vectorization is a mapping that converts $\mathcal{T}$ into a vector $\mathbf{t} \in \mathbb{R}^D$, where $D = d_1 \times d_2 \times \cdots \times d_N$. This involves flattening $\mathcal{T}$ by systematically enumerating its elements, typically in lexicographic or row-major order. Each element $\mathcal{T}_{i_1,i_2,\ldots,i_N}$ is mapped to the corresponding element in the vector $\mathbf{t}$, such that $\mathbf{t}[k] = \mathcal{T}_{i_1,i_2,\ldots,i_N}$, where k is computed as $k = 1 + \sum_{n=1}^{N}(i_n - 1)\left(\prod_{m=n+1}^{N} d_m\right)$ for $1 \leq i_n \leq d_n$.

For the raw data approach, at each time point t, the independent variable data is represented as $\mathbf{X}_t \in \mathbb{R}^{M \times N}$. After vectorization, this becomes a vector $\mathbf{z}_{1,t} \in \mathbb{R}^{P_1 \times 1}$, where $P_1 = M \times N$. For the Tucker decomposition approach, assuming the factor dimensions are $d_{2,r} \times d_{2,c}$, the independent variable data at time point t, after vectorizing the factors, is represented as $\mathbf{z}_{2,t} \in \mathbb{R}^{P_2 \times 1}$, where $P_2 = d_{2,r} \times d_{2,c}$. In the combined approach using main effect decomposition and Tucker decomposition, we assume that the row effect $\boldsymbol{\alpha}_t \in \mathbb{R}^{M \times 1}$, the column effect $\boldsymbol{\beta}_t \in \mathbb{R}^{N \times 1}$, and the core latent factor $\mathbf{F}_t \in \mathbb{R}^{d_{3,r} \times d_{3,c}}$ have already been determined. We first vectorize the core latent factor $\mathbf{F}_t$ to obtain the vector $\mathbf{f}_t = (f_{t,11}, f_{t,12}, \ldots, f_{t,21}, f_{t,22}, \ldots, f_{t,d_{3,r}d_{3,c}})^T$. The prediction variables at time t can then be constructed as $(\boldsymbol{\alpha}_t^T, \boldsymbol{\beta}_t^T, \mathbf{f}_t^T)^T = (\alpha_{t,1}, \ldots, \alpha_{t,M}, \beta_{t,1}, \ldots, \beta_{t,N}, f_{t,11}, \ldots, f_{t,d_{3,r}d_{3,c}})^T$, forming a vector of length $M + N + d_{3,r} \times d_{3,c}$. For convenience, this can be rewritten as $\mathbf{z}_{3,t} \in \mathbb{R}^{P_3 \times 1}$, where $P_3 = M + N + d_{3,r} \times d_{3,c}$. Finally, for all predictors from 1 to T, the data is represented in matrix form as $\mathbf{Z} = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_T)^T$, where each $\mathbf{z}_t$ (with $t \in \{1, 2, 3, \ldots, T\}$) can be $\mathbf{z}_{1,t}$, $\mathbf{z}_{2,t}$, or $\mathbf{z}_{3,t}$.

After determining the form of the predictors, we can perform constrained regression in a generalized form:

$$\min_{\mathbf{c}} \left\{ ||\mathbf{y} - \mathbf{Z}\mathbf{c}||^2 + \sum_{p \in P} p_\delta(c_p) \right\} \tag{3.14}$$

where $\mathbf{c} = (c_1, c_2, \ldots, c_P)$ represents the regression coefficients and p_δ is a scalar penalty function. In this research, we apply Lasso regression, which employs least absolute shrinkage

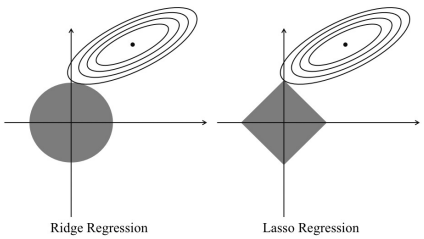


Figure 5: Constrained regression illustration

and selection operation to exclude variables that are not useful for prediction. Specifically, we use $p_\delta(c) = \delta|c|$, which has the ability to penalize a coefficient to exactly 0 (as shown on the right of Figure 5), effectively performing variable selection. We do not use Ridge regression, where $p_\delta(c) = \delta||c||^2$, because it generally includes all covariates in the regression, as no coefficient in $\hat{\mathbf{c}}$ will be set to exactly 0 (as shown on the left of Figure 5). The penalty coefficient δ in the function $p_\delta(c)$ is treated as a hyperparameter, and its value is determined through cross-validation.

To compare the effects of the three different approaches—using raw data, Tucker decomposition, and the combination of main effect decomposition and Tucker decomposition—we adopt mean square error (MSE) as the performance measure, expressed as:

$$MSE = \sum_{t=1}^{T} \frac{(y_t - \hat{y}_t)^2}{T} \tag{3.15}$$

where y_t is the true response value at time t and $\hat{y}_t$ is the fitted value.

4 Simulation Analysis

To date, we have employed traditional Tucker decomposition to develop both the main effect decomposition and its combination, with the goal of extracting the time-varying grand mean, row effects, and column effects to minimize noise and interference in the core factor. In this section, we will conduct simulations to evaluate and compare the performance of Tucker decomposition applied alone versus its integration with main effect decomposition. This comparison will highlight the advantages and limitations of each approach in handling the tensor data.

To ensure a comprehensive comparison, we divide the dataset formation process into two distinct paths, referred to as Path A and Path B, as defined in Equation 4.1 and Equation 4.2, respectively:

$$\mathbf{A}_t := \mathbf{R}\mathbf{F}_t\mathbf{C}^T + \mathbf{e}_t \tag{4.1}$$

$$\mathbf{B}_t := u_t\mathbf{1}_M\mathbf{1}_N^T + \mathbf{a}_t\mathbf{1}_N^T + \mathbf{1}_M\mathbf{b}_t^T + \mathbf{A}_t \tag{4.2}$$

Initially, for both paths, we simulate the row loading matrix $\mathbf{R}$, the column loading matrix $\mathbf{C}$, the core factor $\mathbf{F}_t$ for time t, and the noise term $\mathbf{e}_t$. The primary distinction between the paths lies in how we handle the simulated tensor data: in Path A, the simulated row loading matrix, column loading matrix, and core factor are combined directly with the noise term. In contrast, Path B involves an additional step where we simulate the row main effect $\mathbf{a}_t$ and the column main effect $\mathbf{b}_t$, incorporating these into the tensor data derived from Path A. This design allows us to create two datasets: $\mathbf{B}_t$, which clearly incorporates row and column effects, and $\mathbf{A}_t$, where such effects are less apparent.

To simulate tensor data with dimensions $100 \times 100 \times 100$, we generate $\mathbf{A}_t \in \mathbb{R}^{100\times100\times100}$ and $\mathbf{B}_t \in \mathbb{R}^{100\times100\times100}$, with the core factor $\mathbf{F}_t \in \mathbb{R}^{5\times5}$ assumed to have dimensions 5×5 at each time point t, resulting in an overall core factor tensor of size $5 \times 5 \times 100$. The elements of $\mathbf{F}_t$ are drawn from a standard normal distribution using Python. For the row loading matrix $\mathbf{R} \in \mathbb{R}^{100\times5}$ and the column loading matrix $\mathbf{C} \in \mathbb{R}^{100\times5}$, elements are similarly simulated from a standard normal distribution, but we apply a demeaning process to each column to meet the constraints specified in Equation 3.3. The row main effect $\mathbf{a}_t \in \mathbb{R}^{100\times1}$ and the column main effect $\mathbf{b}_t \in \mathbb{R}^{100\times1}$ are also simulated from a standard normal distribution and are demeaned accordingly. The same procedures are applied to the error term $\mathbf{e}_t$.

After obtaining the simulated tensor data $\mathbf{A}_t$ and $\mathbf{B}_t$ from Equation 4.1 and Equation 4.2, we apply the decomposition formulas provided in Equation 2.5, Equation 3.2, and Equation 3.11. This allows us to derive the core factor using both Tucker decomposition alone and

a combined approach of main effect decomposition followed by Tucker decomposition. For clarity, we refer to the method employing only main effect decomposition as Method 1 and the method combining main effect decomposition with Tucker decomposition as Method 2. Applying Method 1 and Method 2 to $\mathbf{A}_t$ and $\mathbf{B}_t$ yields four types of estimates, labeled Type A1, Type A2, Type B1, and Type B2. Specifically, Type A1 and Type B1 estimates include the row and column loading matrices and the core factor term, while Type A2 and Type B2 include additional estimates such as row and column main effects and grand means. We then assemble these estimates to reconstruct the tensor data, resulting in $\hat{\mathbf{A}}_{t,1}$, $\hat{\mathbf{A}}_{t,2}$, $\hat{\mathbf{B}}_{t,1}$, and $\hat{\mathbf{B}}_{t,2}$. To evaluate the performance of the methods, we compare these estimates to the true values by calculating the relative mean squared errors (ReMSE) as defined by Lam and Cen (2024):

$$ReMSE := \frac{\sum_{t=1}^{T} ||\mathbf{X}_t - \hat{\mathbf{X}}_t||^2}{\sum_{t=1}^{T} ||\mathbf{X}_t||^2} \tag{4.3}$$

where a smaller ReMSE indicates a better estimate. The ReMSE values for the four types of estimates are denoted as $ReMSE_{A1}$, $ReMSE_{A2}$, $ReMSE_{B1}$, and $ReMSE_{B2}$.

According to the data formation method and the two decomposition approaches for each dataset, along with the $ReMSE$ calculation process, we repeat the simulation 1000 times to analyze the four types of $ReMSE$ series obtained. The results are summarized in the box plot shown in Figure 6. This figure illustrates the distribution of $ReMSE$ values for each method

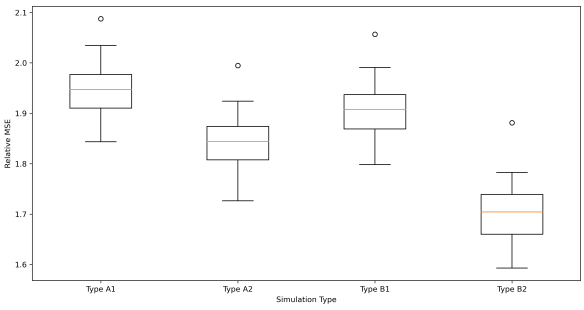


Figure 6: Relative MSE plot

and data path. As depicted, Method 2 consistently outperforms Method 1 across different types of simulated tensor data. Specifically, Method 2 demonstrates superior accuracy in estimating the true values, with particularly pronounced advantages when the data exhibits distinct row and column main effects. In such cases, Method 2's estimates align more closely with the true values, highlighting its effectiveness in capturing the underlying structure of the data. This comparative analysis underscores that integrating main effect decomposition with Tucker decomposition (Method 2) enhances the precision of the estimates and reduces the error compared to using main effect decomposition alone (Method 1). Thus, the simulation

results validate the benefits of incorporating main effect decomposition to improve overall estimation accuracy.

5 Empirical Study

In this section, we will apply the new method introduced in Section 3 Methodology and demonstrate its advantages by comparing it with the research findings and traditional methods outlined in Section 2 Literature Review. We will start by detailing the benchmark data used in this study and constructing relevant features that have proven to be effective. Specifically, to build the independent variables for the forecast model, we first collect individual stock data, while for the dependent variables, we gather information related to the index. We will then evaluate the effectiveness of the main effect decomposition based on the prediction results. Additionally, this study utilizes the Wharton Research Data Services (WRDS) database and conducts data analysis using Python and SQL.

5.1 Data collection and processing

For benchmakring process, we choose the STOXX 50 stock index, which represents the performance of the 50 largest companies across supersectors in nine Eurozone countries, including France, Germany, Ireland, and the Netherlands. This index, part of the STOXX blue-chip family, has a fixed number of components and captures approximately 60% of the free-float market cap of the EURO STOXX total market index. Moreover, it is one of the most liquid indices in the Eurozone. Figure 7 illustrates the price trend of the STOXX 50

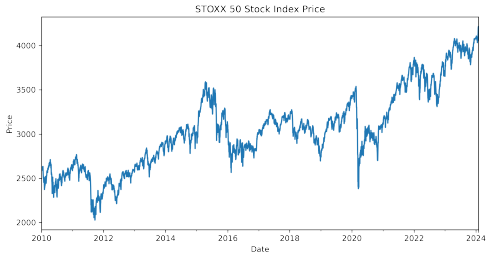


Figure 7: Index price illustration

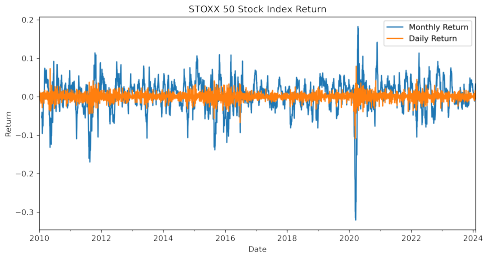


Figure 8: Index return illustration

stock index from 2010 to the end of 2023. Over the long term, the index demonstrates a clear upward trajectory, though it has experienced several notable declines. Significant downturns in the STOXX 50 index during this period can be attributed to three major factors: the European debt crisis in 2011, which intensified debt issues across many European countries and impacted the market; the Chinese stock market crash in 2015, which, despite primarily affecting Asian markets, also had global repercussions; and the COVID-19 pandemic, which delivered a substantial shock to the global economy in early 2020. These events highlight the STOXX 50 index's ability to reflect market conditions and economic environments, under-

scoring the importance of analyzing and studying it.

After determining the STOXX 50 stock index as the research target, the next goal is to select the time period for the study. The STOXX 50 Index was launched on February 26, 1998, by STOXX Limited, a subsidiary of Deutsche Börse Group. Over time, the STOXX 50 stock index's rules for its selected constituents have changed accordingly. The most recent change occurred in September 2020. Effective from the September 2020 review, the updated selection

Table 2: Index constituent selection rule

EURO STOXX 50 Index Component Selection	
Step 1	Rank stocks by free-float market capitalization for each of the 20 EURO STOXX Supersector indices. Add stocks to the selection list until coverage is close to but below 60% of the free-float market cap of the corresponding EURO STOXX TMI Supersector Index. Include all EURO STOXX 50 Index stocks.
Step 2	Rank the stocks on the selection list by free-float market capitalization to produce the final index selection list.
Step 3	Select the top 40 stocks from the final list. Choose the remaining 10 stocks from those ranked 41 to 60. If fewer than 50 stocks are selected, include additional top stocks until the total reaches 50.

rule is outlined in Table 2. Compared to the previous rule in place until the September 2020 review, the number of EURO STOXX Supersector indices increased from 19 to 20, marking a significant adjustment in recent years. To ensure the stability of the STOXX 50 Index as a benchmark, we will use data following these rule changes. Specifically, the index undergoes an annual review each September, with the review cut-off date being the last trading day of August. For consistency and to align with the update schedule of the WRDS database, our study covers all trading days from September 1, 2020, to December 31, 2023. In the end, we successfully acquired data from 838 time points.

After selecting the time period, the next step is to select the target portfolio. We select all the constituents of the STOXX 50 index as of September 2020 from Index Constituents pattern in WRDS, and accordingly select the stocks that have been included in the constituents through the end of 2023. The results show that the stock composition of the STOXX 50 (e.g., NESTLE SA/AG, LVMH MOET HENNESSY LOUIS V, ASML HOLDING NV) has not changed during the selected time period, which confirms the stability of the index based on the stock selection rules in September 2020.

After determining the research objective and time interval, it is essential to clarify the independent and dependent variables in the model. For the independent variable, we treat different stocks at each point in time as individual observations, collect data, and construct factors. Generally, the data consists of trading information from daily market updates. Given

Table 3: Variable illustration

Name	Definition
prccd	This item represents the close price of the day for the security
cshoc	This item represents the current number of common/ordinary shares outstanding for a particular issue of stock
cshtrd	This item represents the number of shares traded for companies listed on national stock exchanges and over-the-counter companies in the NASDAQ system
curcdd	This item identifies the currency in which daily prices are available
ajexdi	This item provides adjustment factors specifically for company issues
trfd	This item represents the daily multiplication factor for calculating the total return to shareholders of any time period
exratd_togbp	This is the actual currency translation rate from native currency to GBP

the short time period, using financial data would result in lower frequency; therefore, this article focuses on daily market trading data for analysis. For the 50 stocks in the target portfolio, we select relevant variables (detailed in Table 3) from the Exchange Rates and Security Daily patterns in WRDS. Several fundamental operations are then performed on the data. First, we convert the daily closing prices of all stocks into GBP based on the day's exchange rate, expressed as $\texttt{prccd_gbp} = \texttt{prccd} \times \texttt{exratd_togdp}$. To adjust for corporate actions like stock splits or dividends, we calculate the adjusted close price using $\texttt{adj_prccd_gbp} = \frac{\texttt{prccd_gbp}}{\texttt{ajexdi}}$. For returns from t_1 to t_2, considering cash distributions, dividend reinvestment, and compounding effects, we use $\frac{\texttt{adj_prccd_gbp}_{t_2} \times \texttt{trfd}_{t_2}}{\texttt{adj_prccd_gbp}_{t_1} \times \texttt{trfd}_{t_1}} - 1$. After this initial data processing, the specific construction of factor variables as predictors is described in Section 5.2.

Next, we consider the response variable in the forecasting model. The return calculated from the STOXX 50 index price will be used as the dependent variable. All data are collected from the Index Daily Prices pattern in WRDS. In Figure 8, the monthly (rolling over 20 trading days) and daily returns of the index from the beginning of 2010 to the end of 2023 are shown. Rounded to two decimal places, their average and unconditional volatility over this period were -0.05 and 20.86 for the monthly returns, and 0.00 and 0.01 for the daily returns, respectively. It can be observed that the monthly returns exhibit more significant variation over time.

In conclusion, we will use the factors at time t to predict the index return one month (20 trading days) after t.

5.2 Feature construction

After collecting and computing the basic market data for each stock in the target portfolio, the next step is to construct the corresponding factor independent variables to enable

accurate forecasting. Specifically, the factors can be categorized into six groups: size, volatility, contrarian, momentum, turnover rate, and cost of illiquidity. It should be noted that in the subscript of the variables introduced below, i represents the i-th stock, and t denotes the point in time.

Size. Banz (1981) demonstrated that a company's market value is inversely related to its rate of return. Building on this, Fama and French (1992) highlighted the significance of the size factor by incorporating it into their three-factor model. In this study, we will use the free-float market cap (denoted as `mkt_cap`) as one of the features, calculated as $\texttt{cshoc}_{t,i} \times \texttt{prccd_gbp}_{t,i}$ where we can use adjusted or non-adjusted daily close price in GBP because in most cases, any adjustments affect both numbers and cancel out. Following Lewellen (2014), we will apply a logarithmic transformation to this market value as $log(\texttt{mkt_cap}_{t,i})$. Additionally, consistent with the author's findings, we will measure the growth in shares outstanding over the past thirty-six months as another measure of size as $\texttt{cshoc_grow}_{t,i} = \frac{\texttt{cshoc}_{t,i}}{\texttt{cshoc}_{t-750,i}}$ where we use 250 trading days to represent a year and also take its logarithmic as $log(\texttt{cshoc_grow}_{t,i})$.

Volatility. Moreira and Muir (2017) introduced a well-known volatility management portfolio strategy, demonstrating that the excess returns generated by their strategies are predominantly driven by volatility. Consequently, we also incorporate a stock's past volatility as an independent variable in our analysis. Specifically, following Liu et al. (2019), we use two volatility-related variables: (i) One-month volatility (denoted as `vol_std20d`), calculated as the standard deviation of daily returns over the past 20 trading days; and (ii) MAX (denoted as `vol_max20d`), which represents the highest daily return observed over the past 20 trading days.

Contrarian. De Bondt and Thaler (1985) discovered that over a certain period, selling stocks that had performed well in the past and buying stocks that had performed poorly resulted in the latter outperforming the market after being held for a certain period. Conversely, the previously well-performing stocks underperformed the market in the future, leading to the proposal of the contrarian effect. Empirical studies by Jegadeesh (1990), Lo and MacKinlay (1990), and Lehmann (1990) found that the contrarian effect in stock prices is significant in the short term, specifically when both the holding and observation periods are within one month. However, when these periods extend from one to twelve months, no significant contrarian effect is observed. Therefore, for the reversal factor in this research, we define the reversal variable (denoted as `rev`) of each stock in month t as the return of the same stock in the previous month (past 20 trading days).

Momentum. Jegadeesh and Titman (1993) document that U.S. stocks that perform the best (worst) over a three to 12-month period tend to continue to perform well (poorly) over the subsequent three to 12 months. In a follow-up study, Jegadeesh and Titman (2001) also show that momentum strategies continued to be profitable. Griffin et al. (2003), Chui et al. (2010) and Asness et al. (2013) find that this strategy has been profitable in most major markets throughout the world, including 12 European countries proven by Rouwenhorst (1998). Therefore, in this article, we will adopt the momentum factor as a key predictor. For a stock at the end of month t, we use its cumulative earnings over the previous one, two, three, and four quarters as the momentum factor (denoted as `mom_1q`, `mom_2q`, `mom_3q`, and `mom_4q` respectively). Notably, as mentioned earlier, the returns from the most recent month can induce a reversal effect. To account for this, when calculating cumulative returns, we exclude the returns of the latest month. For instance, if one quarter is taken as the observation period, the corresponding momentum factor should be the cumulative returns from the end of month $t-3$ to the end of month $t-1$.

Turnover rate. Amihud and Mendelson (1986) pioneered the liquidity premium theory, one of the earliest studies exploring how stock liquidity influences expected returns. Subsequent research on turnover effects has largely built upon their work. Haugen and Baker (1996) observed a significant negative correlation between expected stock returns and turnover in studies conducted across several countries, including the UK, Germany, and France. This negative relationship between turnover and stock returns has shown a degree of stability across different periods and countries. Accordingly, in this study, we also include turnover rate as one of the predictors. Daily turnover rate is calculated as the share trading volume divided by the total shares outstanding with formula $\frac{\texttt{cshtrd}}{\texttt{cshoc}}$. Following Liu et al. (2019), we consider two variables in this context: (i) Twelve-month turnover (denoted as `tor_avg250d`), defined as the average daily share turnover over the past 250 days; (ii) One-month turnover (`tor_avg20d`), defined as the average daily share turnover over the past 20 days; and (iii) One-month abnormal turnover (denoted `tor_ab`), calculated as the ratio of the average daily turnover over the past 20 days to the average daily turnover over the past 250 days.

Cost of illiquidity. Demsetz (1968) firtly discussed how transaction costs affect the formation process of prices from the perspective of market microstructure. Therefore, this research will use two features to measure the cost of illiquidity. The first one follows the research proposed by Amihud (2002), where he conduct a stock's illiquidity measure for day t calculated as $\texttt{illiq}_{t,i} = \frac{|\texttt{ret}_{t,i}|}{\texttt{cshtrd}_{t,i}}$, where $|\texttt{ret}_{t,i}|$ is the i-th stock's absolute return on day t. At the end of each month, we compute a stock's average daily illiquidity over the past 20

trading days (denoted as `illiq_avg20d`) as one of the factors.

Up to this point, we have examined thirteen factors across six broad categories. For our data structure at time point t, we create a matrix with stocks as rows and factors as columns, resulting in a dimension of 50×13. With a total of 838 time points, our dataset forms a $50 \times 13 \times 838$ cube. For the matrix-valued data at each time point, we apply main effect decomposition, which involves computing the eigenvalues and eigenvectors of the matrix. We then calculate the correlation coefficient matrix for different factors across various stocks at each cross-section and average these correlations across the time dimension for display (as shown in Table 4). The result reveals several patterns: factors within the same category

Table 4: Cross-sectional factor correlation matrix

	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**
1. `illiq_avg20d`	1.0	0.4	-0.1	-0.5	-0.4	-0.1	0.0	0.0	0.1	0.1	0.0	-0.0	-0.0
2. `log(mkt_cap)`		1.0	0.1	-0.6	-0.6	-0.1	-0.2	-0.1	0.2	0.1	0.1	0.1	0.0
3. `log(cshoc_grow)`			1.0	-0.0	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.0	-0.0
4. `tor_avg250d`				1.0	0.9	-0.0	0.2	0.2	0.0	0.1	0.1	0.0	0.0
5. `tor_avg20d`					1.0	0.4	0.3	0.2	-0.0	-0.0	0.0	-0.0	-0.0
6. `tor_ab`						1.0	0.3	0.2	-0.1	-0.1	-0.1	-0.1	-0.1
7. `vol_std20d`							1.0	0.7	0.0	0.0	0.0	-0.0	-0.1
8. `vol_max20d`								1.0	-0.1	-0.1	-0.1	-0.1	0.3
9. `mom_4q`									1.0	0.8	0.6	0.4	-0.0
10. `mom_3q`										1.0	0.7	0.5	-0.0
11. `mom_2q`											1.0	0.6	-0.1
12. `mom_1q`												1.0	-0.1
13. `rev`													1.0

exhibit high correlations, such as between `tor_avg250d` and `tor_avg20d`, `vol_std20d` and `vol_max20d`, and among `mom_4q`, `mom_3q`, `mom_2q`, and `mom_1q`. Liquidity-related factors, such as cost of illiquidity and turnover rate, show a strong negative correlation, reflecting their opposing directions of liquidity. Additionally, a significant negative correlation is observed between market value and turnover rate factors. The turnover rate indicators, calculated from `cshoc` and `cshtrd`, have average values of approximately 5916.28 and 17.56, respectively, indicating a large difference in magnitude. This suggests that the turnover rate index primarily depends on share outstanding, which rationalizes the observed negative correlation between market value and turnover rate indicators.

5.3 Result and discussion

Based on the previous sections, we obtain data in cubic form, denoted as $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{50\times13\times838}$. Specifically, for each time point $t \in \{1, 2, 3, \ldots, 838\}$, we have $\mathbf{X}_t$ in matrix form with dimensions 50×13. Using the timeline as a dividing line, we partition the overall dataset into a training set and a test set with an 8:2 ratio, denoted as $\boldsymbol{\mathcal{X}}_{train} \in \mathbb{R}^{50\times13\times670}$ and

$\boldsymbol{\mathcal{X}}_{test} \in \mathbb{R}^{50\times13\times138}$, respectively. We fit the data from the training set and make predictions using the test set to evaluate the advantages and disadvantages of the three existing models. These three models represent different approaches for handling the independent variables.

The first approach involves no preprocessing; instead, each time point $\mathbf{X}_t$ is directly vectorized, and the corresponding regression is performed on the dependent variable. This approach is referred to as Model A. For the training phase, the vectorized independent variables in Model A are denoted as $\mathbf{z}_{A,t} \in \mathbb{R}^{650\times1}$, and the response variable is $y_{t+1} \in \mathbb{R}$, where $t \in \{1, 2, 3, \ldots, 670\}$. Similarly, for the testing phase, the vectorized independent variables are $\mathbf{z}_{A,t} \in \mathbb{R}^{650\times1}$, and the response variable is $y_{t+1} \in \mathbb{R}$, where $t \in \{671, 672, 673, \ldots, 838\}$.

The second model involves a direct Tucker decomposition of $\boldsymbol{\mathcal{X}}_{train}$ and $\boldsymbol{\mathcal{X}}_{test}$, referred to as Model B. Initially, we need to determine the number of factors for the Tucker decomposition of $\boldsymbol{\mathcal{X}}_{train}$. Following the steps outlined in Equation 3.12 and Equation 3.13, we calculate the eigenvalues of the sample row and column covariance matrices, sort them in descending order, and compute the eigenvalue-ratios, as shown in the following plots. Figure 9 and Figure 10

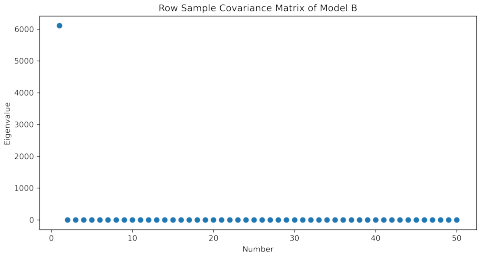


Figure 9: Model B row eigenvalue

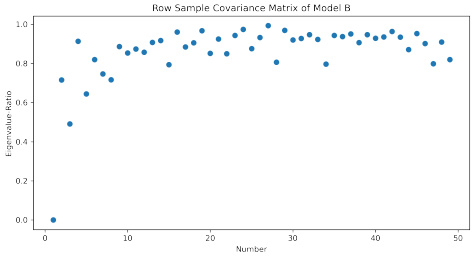


Figure 10: Model B row eigenvalue-ratio

show the results for the sample row covariance matrix in Model B. The largest eigenvalue of the sample row covariance matrix explains over 99.99% of the total variation, and the smallest eigenvalue-ratio is observed when the row factor number is set to 1. Similarly,

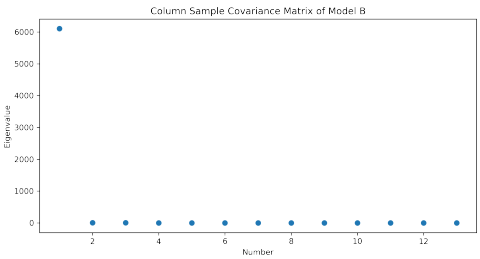


Figure 11: Model B column eigenvalue

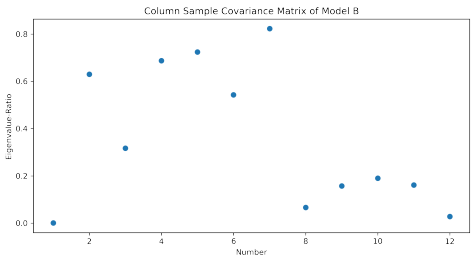


Figure 12: Model B column eigenvalue-ratio

Figure 11 and Figure 12 display the eigenvalues and eigenvalue-ratios for the column sample covariance matrix. The largest eigenvalue again dominates and shows the smallest eigenvalue-ratio. Therefore, the number of column factors should also be set to 1. Overall, the factor

dimension after applying Model B is 1×1. Consequently, the independent variable at time t in Model B can be expressed as $z_{B,t}$, a scalar where $t \in \{1, 2, 3, \ldots, 670\}$ for the training set and $t \in \{671, 672, 673, \ldots, 838\}$ for the test set.

Based on the dimensionality of the factor term from Model B, we observe that the variation between each row (stocks) and each column (features) is minimal, as only one factor can explain all the data. This suggests that there may be a common pattern that accounts for a significant portion of the variation across different stocks and features, making individual differences appear very small. To address this issue, we will apply Model C, which introduces a new main effect decomposition compared to Model B, to better capture the underlying patterns and improve upon the limitations identified in Model B.

During the processes for Model C, we first apply the new method shown in Equation 3.2 to each $\mathbf{X}_t$ in $\boldsymbol{\mathcal{X}}_{train}$. For the resulting interaction term $\boldsymbol{\gamma}_{train} \in \mathbb{R}^{50\times13\times670}$, we then perform a Tucker decomposition as shown in Equation 3.11 to generate the factor term. As with Model B, we determine the dimensionality of the factor term by analyzing the eigenvalues of the row and column sample covariance matrices of $\boldsymbol{\gamma}_{train}$. Figure 3.13 illustrates the sorted

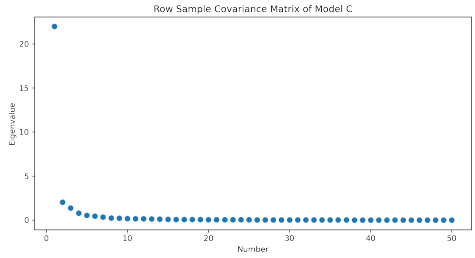


Figure 13: Model C row eigenvalue

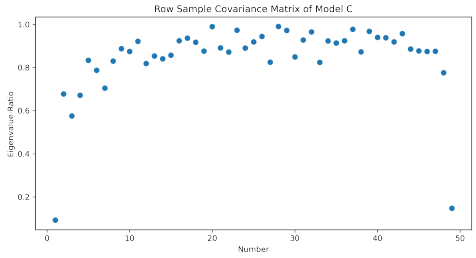


Figure 14: Model C row eigenvalue-ratio

eigenvalues calculated using the row sample covariance matrix. According to Equation 3.12 and Equation 3.13, the number of core factors in Model C could be determined as 1, as it yields the lowest eigenvalue-ratio shown in Figure 3.14. However, upon examining the explanatory power of this first factor, we find that it accounts for only 74.22% of the total variation. Drawing on the approach proposed by Martellini et al. (2003) for identifying implicit factor models in the fixed-income domain, we also consider the number of factors that can account for more than 85% of the total variation. In this case, the first three factors contribute 85.7%, which meets our criterion. Therefore, when determining the number of row factors for the core factor in Model C, we consider both one and three factors. Figure 15 and Figure 16 illustrate the eigenvalues and eigenvalue-ratios computed from the column sample covariance matrix for Model C. The results clearly indicate that eight factors yield the minimum eigenvalue-ratio, with the first eight eigenvalues accounting for approximately

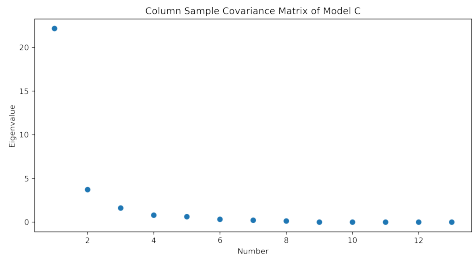


Figure 15: Model C column eigenvalue

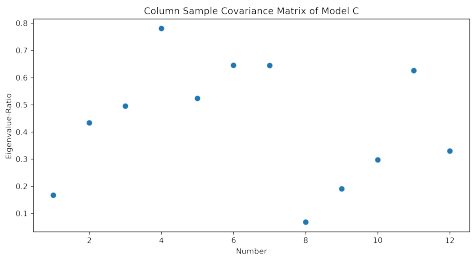


Figure 16: Model C column eigenvalue-ratio

99.99% of the total variation. Thus, eight factors are identified as the appropriate number of column factors for Model C. We have already identified two alternative dimensions for the core factors: one and eight, and three and eight. To maintain clarity in the following analysis, we subdivide Model C into two versions: Model C1, corresponding to the first factor dimension, and Model C2, corresponding to the second. Consequently, in the analysis of Model C1, the vectorized independent variable is represented as $\mathbf{z}_{C1,t} \in \mathbb{R}^{8\times 1}$ with $t \in \{1, 2, 3, \ldots, 670\}$ for the training set and $t \in \{671, 672, 673, \ldots, 838\}$ for the test set. For Model C2, the independent variable is represented as $\mathbf{z}_{C2,t} \in \mathbb{R}^{24\times 1}$ with $t \in \{1, 2, 3, \ldots, 670\}$ for the training set and $t \in \{671, 672, 673, \ldots, 838\}$ for the test set.

The results from Model C1 and C2 show significant variation in factor dimensions following the application of main effect decomposition to extract the row and column main effects. This variation is most noticeable in the factors associated with column numbers, where differences between features become more distinct after adjusting for their common main effects. Notably, the factors can be classified into eight distinct groups. However, in the stock dimension, the main effect decomposition did not perform as expected, resulting in increased differentiation among stocks from different industries. This may be because stocks that are part of the STOXX 50 index display prominent characteristics across all dimensions, thereby reducing variability in the row dimensions of the tensor data.

Next, after completing the data processing for Models A, B, C1, and C2, we construct the Lasso regression models using their respective datasets and report the mean squared error (MSE) for both the training and test sets, as defined in Equation 3.15. According to the

Table 5: Model comparison

Model	**Training MSE**	**Test MSE**
A	1.56×10^{-5}	1.44×10^{-2}
B	1.38×10^{-3}	4.50×10^{-3}
C1	3.93×10^{-4}	1.03×10^{-3}
C2	3.33×10^{-4}	3.85×10^{-3}

results in Table 5, the training and test MSEs for Models A, B, C1, and C2 can be compared.

In terms of training MSE, Model A performed the best, which is expected given that it utilizes the most independent variables in the training set. However, this advantage also results in overfitting, as evidenced by the test MSE, where Model A performed the worst. Models C1 and C2 outperformed Model B in both training and test MSEs, highlighting the effectiveness of the newly proposed main effect decomposition in this thesis. When combined with Lasso regression, although the datasets processed by Models C1 and C2 contain more variables, they effectively manage noise, retaining useful information while eliminating redundancy. Between Models C1 and C2, despite both employing a combination of main effect decomposition and Tucker decomposition and yielding similar training and test MSE values, Model C2 slightly outperformed Model C1 on the training set, while Model C1 had a slight edge on the test set. This discrepancy may be due to Model C2 containing slightly more independent variables than Model C1, leading to minor overfitting. Additionally, this suggests that after processing the data with the combination of main effect decomposition and Tucker decomposition, the optimal number of row factors is likely 1, indicating that the variation among different stocks may not be particularly significant.

6 Conclusion

In this section, we will conclude the paper with a summary, outline potential future development directions, and discuss the limitations that need improvement.

6.1 General summary

As we approach the conclusion of this paper, we have focused on the study of latent factor models applied to three-dimensional tensor data. Specifically, we segmented the data along the time axis, obtaining matrix-valued data (i.e., two-dimensional tensor data) at each time point. Based on this, we introduced a novel method called main effect decomposition within the context of matrix-valued Tucker decomposition. This method extracts the time-varying grand mean, as well as the row and column main effects, to capture the common patterns in the data. As a result, the data becomes less noisy and more precise. By combining Tucker decomposition with main effect decomposition, we achieve both dimensionality reduction and enhanced interpretability of the data, leading to clearer structures and more accurate analysis.

From a theoretical perspective, Section 3 Methodology provides a detailed explanation of the derivation process, from main effect decomposition to the traditional Tucker decomposition, and finally to constrained regression. We also list the parameters that need to be estimated in detail.

From a practical standpoint, Sections 4 Simulation Analysis and Section 5 Empirical Study demonstrate the superiority of main effect decomposition. In the former, we used Python to generate random data, applying various data processing methods to verify that main effect decomposition significantly improves the accuracy of data estimation. In the latter, we selected data from the STOXX 50 stock index, a benchmark in the real market, and calculated fundamental and effective factor indicators, as identified in the literature, to predict future index returns. The results indicate that, in addition to using Tucker decomposition for dimensionality reduction to address overfitting, the additional step of main effect decomposition further enhances prediction accuracy.

6.2 Further development

Based on the foundation laid in this paper, there are several directions for further improvement in the future.

First, regarding the selection of factor indicators, we have chosen six fundamental categories in this study: size, volatility, contrarian, momentum, turnover rate, and cost of

illiquidity. To further enhance predictive performance, we could incorporate additional fundamental factors, such as profitability and value factors derived from financial statements. A challenge with these factors is the data frequency. Moreover, with the rapid development of the quantitative finance industry, many complex factors have been proven effective, and they could be considered in future analyses. Lastly, we might also consider integrating ESG factors. In many parts of the world, especially in Europe, ESG scores are crucial indicators of sustainable development and are a focus for governments and nations today. Therefore, incorporating them into model construction could be valuable.

Another area for improvement lies in the construction of the final prediction model. Currently, when constructing and processing independent variables, we vectorize high-dimensional data, which to some extent, compromises the original data structure. In future analyses, we could consider using tensor regression techniques, such as importance sketching created by Zhang et al. (2020), to further enhance prediction accuracy.

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